
AWS → CLOUDFLARE MIGRATION · PROJECT DOCUMENTATION

Architecture Overview

Supporting Document 1 — Before & After, and Traffic Flow

SaaS Edge Layer - AWS CloudFront/WAF/Shield → Cloudflare

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Audience: Client / CTO / Engineering Team

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1. Purpose

This document explains the edge architecture before and after the migration and describes how traffic flows through each setup. It is a companion to the main project documentation and is intended to give the client a clear, side-by-side picture of what changed and why, while making clear that the AWS application stack was left untouched.

2. Before Architecture - AWS Edge Stack

In the original architecture, public traffic passed through a stack of AWS edge and security services before reaching the Application Load Balancer. Each service was configured and billed separately, and each had its own console.

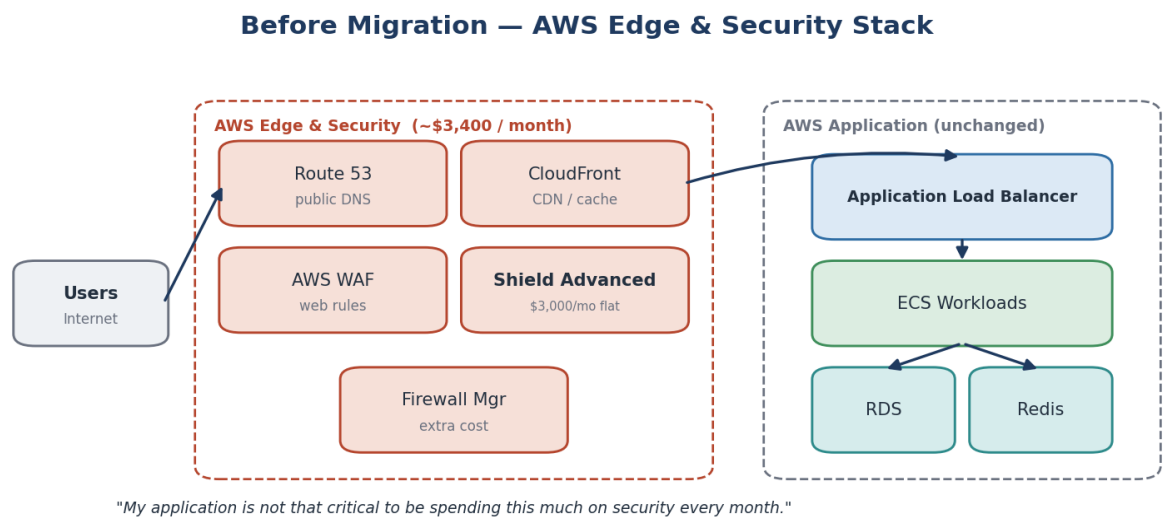


Figure 1 - AWS edge and security stack (before)

2.1 The edge components

Component	Role
Route 53	Public DNS for the domain.
CloudFront	CDN and caching at the edge.
AWS WAF	Web application firewall rules.
Shield Advanced	Managed DDoS protection, \$3,000/month flat.
Firewall Manager	Centralized security policy management.

2.2 The application (unchanged)

Behind the edge, the application ran on a standard AWS stack, the Application Load Balancer fronted the ECS workloads, which used RDS and Redis for data and caching, all inside the client's VPC. None of this was part of the problem, and none of it moved.

- Application Load Balancer (ALB) - entry point to the application.
- ECS workloads - the application services themselves.
- RDS and Redis - database and cache.
- VPC, security groups, and internal networking.

3. After Architecture - Cloudflare + AWS Origin

In the new architecture, Cloudflare sits in front of everything as the public edge. It handles DNS, CDN, WAF, and DDoS protection, then proxies traffic to the AWS ALB as the origin. The application stack is exactly as it was Cloudflare simply replaced the AWS edge layer.

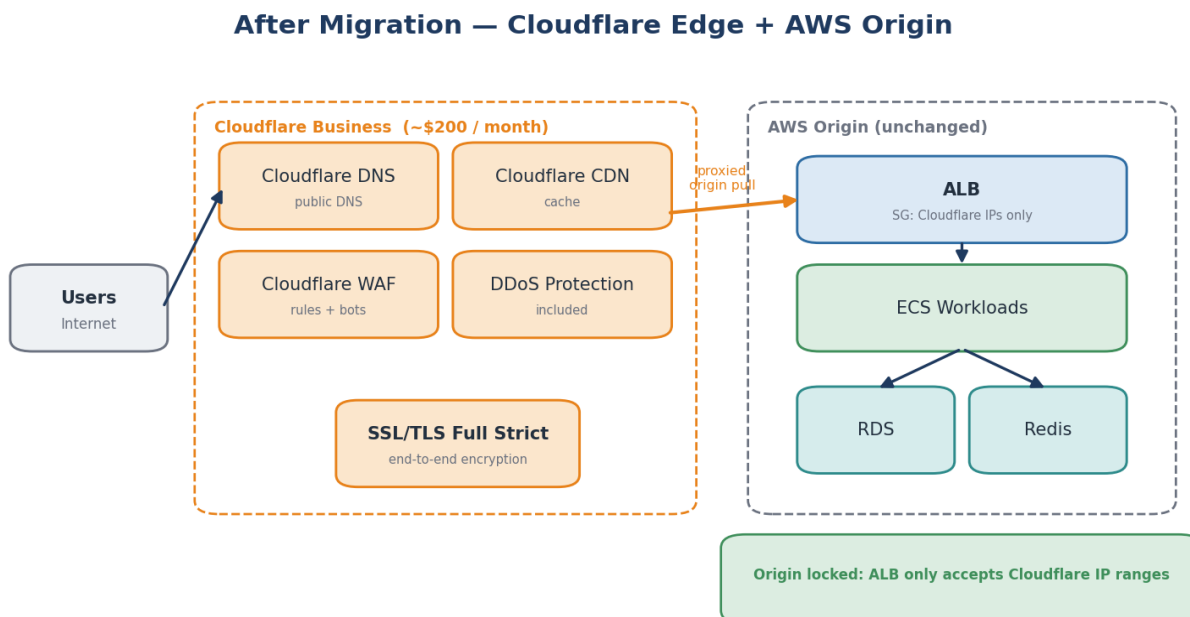


Figure 2 - Cloudflare edge with AWS origin (after)

3.1 The new edge

Cloudflare component	Replaces
Cloudflare DNS	Route 53 public DNS
Cloudflare CDN	CloudFront caching
Cloudflare WAF	AWS WAF rules
Cloudflare DDoS protection	Shield Advanced
SSL/TLS Full Strict	Edge encryption configuration

3.2 The origin (still AWS)

The AWS ALB is now the Cloudflare origin. Its security group was locked down to accept traffic only from Cloudflare IP ranges, so the origin cannot be reached directly. Everything behind the ALB, ECS, RDS, Redis, the VPC is unchanged.

End-to-end encryption

SSL/TLS is set to Full Strict mode, so traffic is encrypted and certificates are validated all the way from the user, through the Cloudflare edge, to the AWS ALB origin.

4. Traffic Flow Comparison

The diagram below places the two request paths side by side. The application destination is the same AWS ALB in both cases — what changed is the edge in front of it and how it is managed.

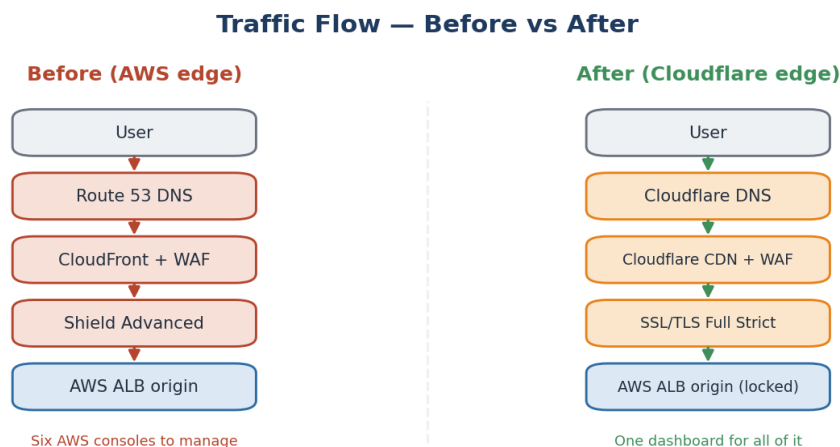


Figure 3 - Request flow, before vs after

4.1 What changed and why

Aspect	Before (AWS edge)	After (Cloudflare)	Why it matters
DNS	Route 53	Cloudflare DNS	Single provider for DNS and edge.
CDN	CloudFront	Cloudflare CDN	Caching managed in one place.
WAF	AWS WAF	Cloudflare WAF	Same rules, simpler management.
DDoS	Shield Advanced (\$3,000)	Included	Right-sized protection, huge saving.
Origin access	Reachable directly	Cloudflare IPs only	No bypassing the edge.
Management	Six consoles	One dashboard	Clear visibility for a non-technical owner.

The headline change

The application path did not change — traffic still ends at the AWS ALB. What changed is that a costly, six-console AWS edge stack was replaced by a single Cloudflare layer that delivers the right level of protection at about 94% lower cost, with the origin safely locked down.

4.2 Summary

- Only the public edge moved; the AWS application stack stayed exactly in place.
- Cloudflare proxies to the AWS ALB, which now only accepts Cloudflare traffic.
- Protection is right-sized to the client's real risk, not an enterprise DDoS profile.
- Everything is managed and observed from one dashboard instead of six AWS consoles.